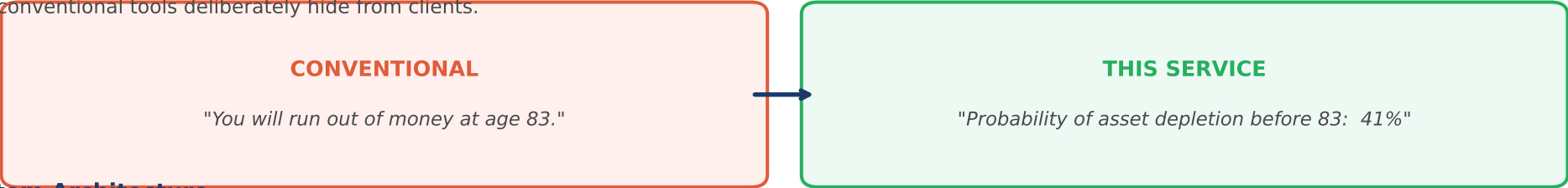


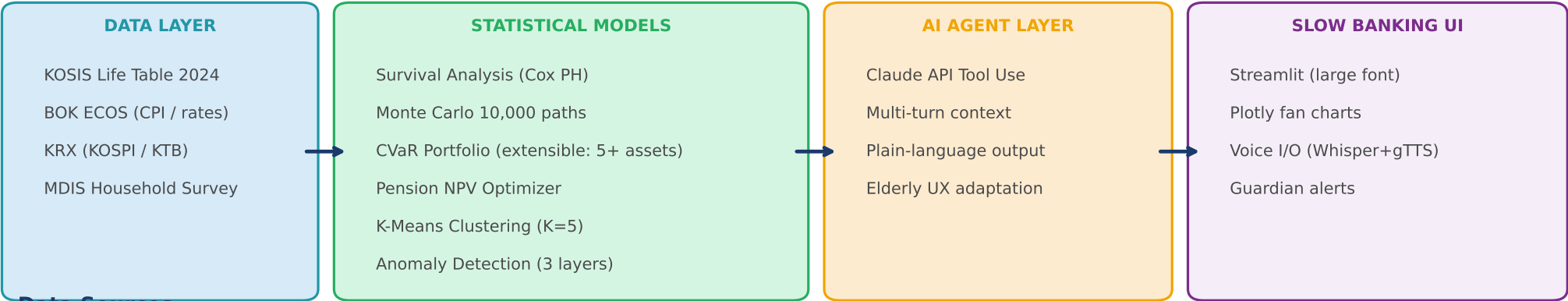
Project Overview

Core Differentiator

Conventional WM services express future wealth as a single deterministic number. LifeLong WM expresses it as a probability distribution— quantifying uncertainty that conventional tools deliberately hide from clients.



System Architecture



Data Sources

Dataset	Source	Period	Usage
National Life Table 2024	Statistics Korea (KOSIS)	2024	Survival Analysis
CPI Monthly Index	Bank of Korea ECOS API	2000-2026	Inflation distribution
Market Interest Rates	Bank of Korea ECOS	2000-2026	Bond yield proxy
KOSPI Daily Price	KRX	2000-2026	Equity return dist.
KTB Bond Total Return	KRX (18 files)	2009-2026	Bond return dist.
Household Finance Survey	MDIS (Statistics Korea)	2025	K-Means clustering

2024 National Life Table (Male)

Age	q(x) death prob.	l(x) survivors	S(x) surv. rate
0	0.00261	100,000	1.0000
40	0.00123	98,125	0.9813
60	0.00645	94,199	0.9420
75	0.02152	73,840	0.7384
80	0.04870	64,373	0.6437
90	0.13441	27,541	0.2754
100+	1.00000	1,153	0.0115

Source: Statistics Korea, 2024 Complete Life Table

Cox Proportional Hazards — Implementation

Step 1 Baseline Survival

$$S_0(t) = \prod_{x=0}^{t-1} (1 - q_x)$$

Cumulative product of life-table death probs.
Male/Female tables applied separately.

Step 2 Conditional on Current Age

$$S(t|a) = S_0(t) / S_0(a)$$

Normalizes survival to current age a.
Eliminates left-truncation bias.

Step 3 Cox PH Adjustment

$$S_{adj}(t) = [S(t|a)]^{HR_{total}}$$

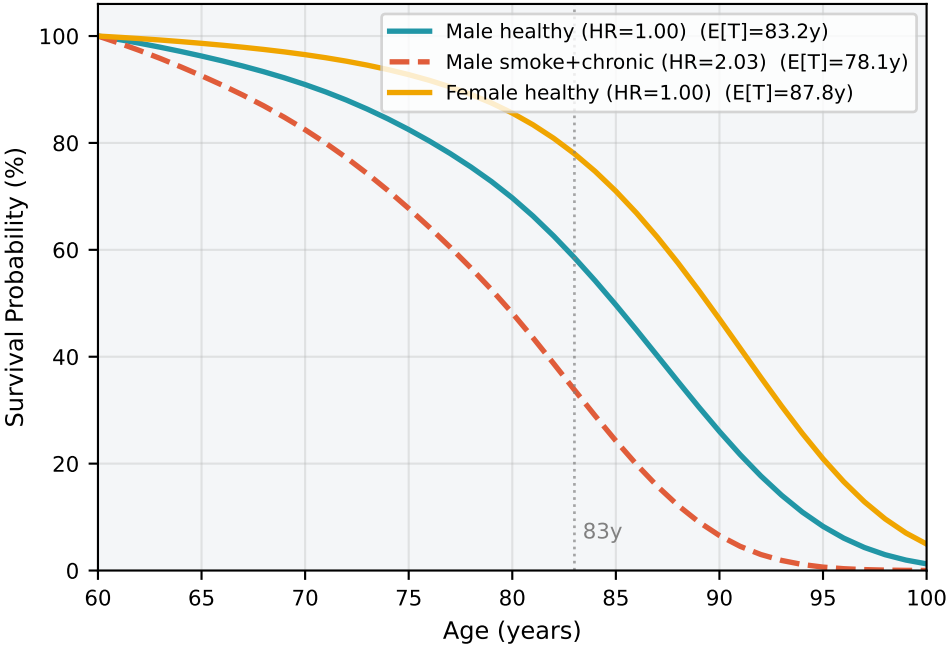
$$HR_{total} = HR_{smoke} \times HR_{chronic} \times HR_{BMI}$$

Smoking 1.50 / Chronic 1.35 / Obese 1.25 / Underweight 1.40

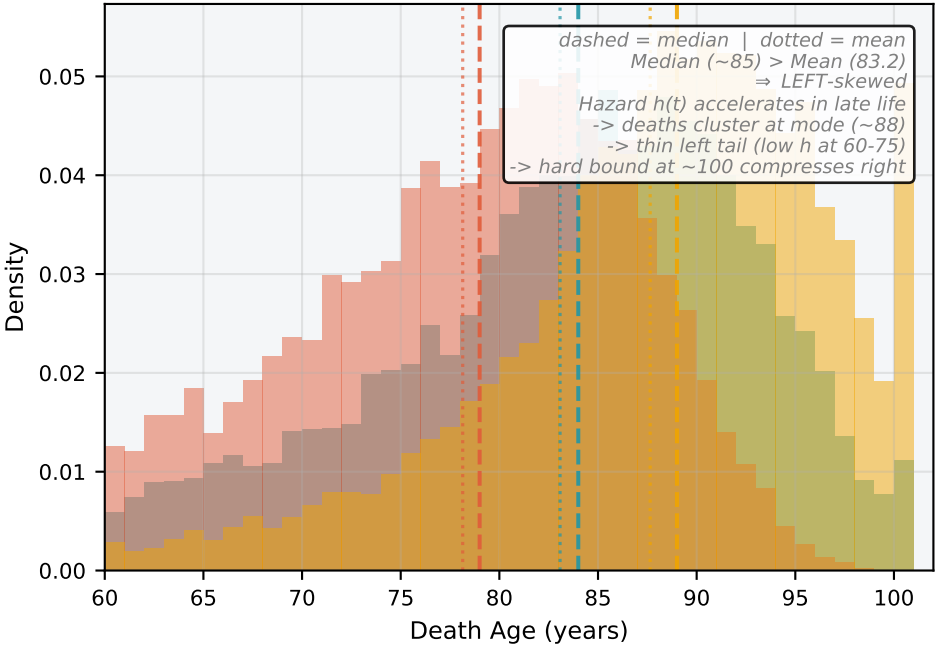
Step 4 Death Age Sampling

$$P(T=t) = S(t) - S(t+1)$$
 (discrete hazard)
Inverse-transform sampling from P(T=t).
Feeds directly into Monte Carlo paths.

Personalized Survival Functions S(t)



Simulated Death Age Distribution (10,000 paths)



Historical Distribution Parameters (KOSPI 2000-2025 / KTB 2009-2025)

Variable	Distribution	Mean	Std Dev
KOSPI equity	Normal	9.56%	28.03%
KTB bond	Normal	2.84%	2.26%
CPI inflation	Normal (clipped)	2.49%	1.34%
Medical shock	Pois x LogNorm	age-varying	500 +/- 300 (10k KRW)
Equity-Bond corr.	Multivariate Normal	-0.025	—

Incomplete year (2026) excluded from parameter estimation.

Simulation Algorithm

```
FOR i = 1 ... N (N = 10,000 paths)
  T_i ~ SurvivalModel.sample_death_age()

  FOR t = 0 ... T_i - 1
    age = current_age + t

    [r_equity, r_bond] ~ MVN([u_s, u_b], Sigma)
    r_port = w * r_equity + (1-w) * r_bond

    pi_t ~ Normal(u_pi, s_pi) clipped [-2%, 15%]

    lambda(age): <70->0.05 70-79->0.10 80+->0.20
    n_t ~ Poisson(lambda(age))
    M_t = n_t * LogNormal(mu_m, sigma_m)

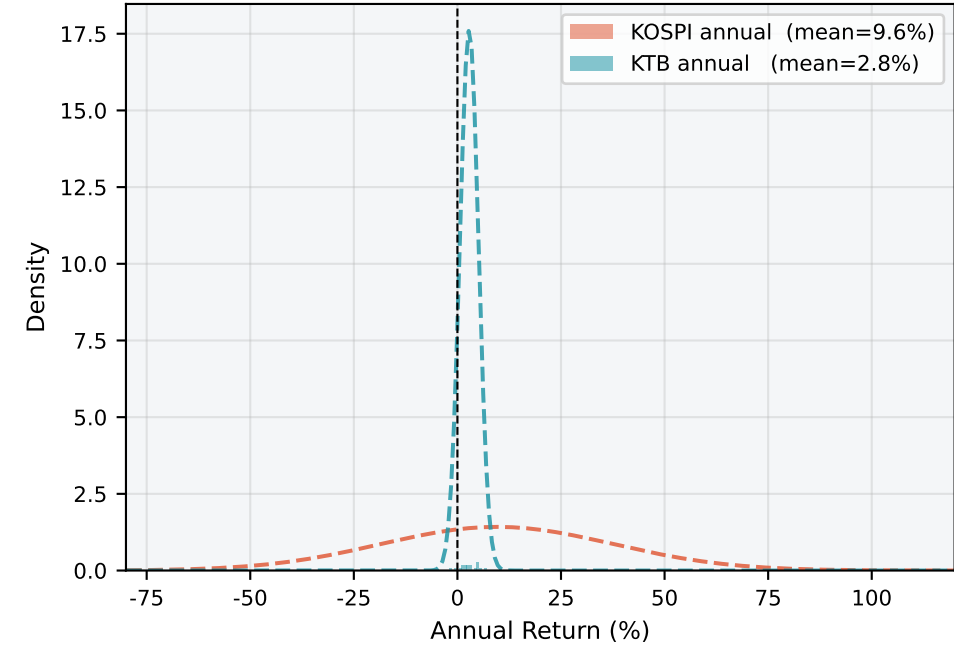
    cum_infl = prod(1 + pi_s, s=0..t)
    E_t = E_0 * cum_infl [real expense]
    P_t = pension * cum_infl [CPI-indexed]
    D_t = E_t + M_t - P_t [net drawdown]

    A_{t+1} = A_t * (1 + r_port) - D_t
    IF A_{t+1} <= 0: record depletion age, stop

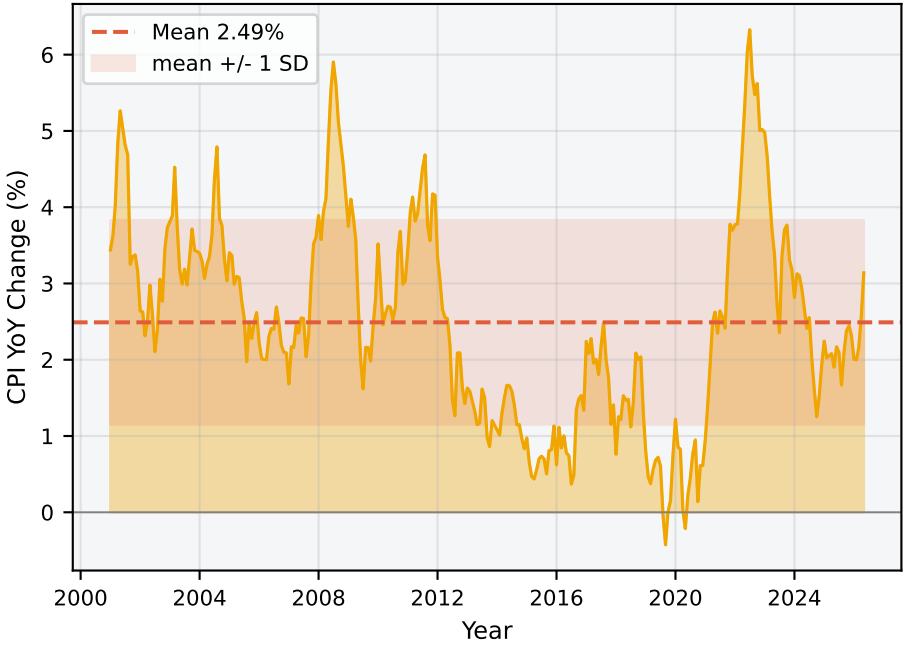
P(depletion) = #{depleted paths} / N
Percentile paths: empirical quantile per age

[Current] inflation sampled i.i.d. per year.
Future: AR(1)/VAR(1) for CPI-rate persistence
(captures inflation clustering across years).
```

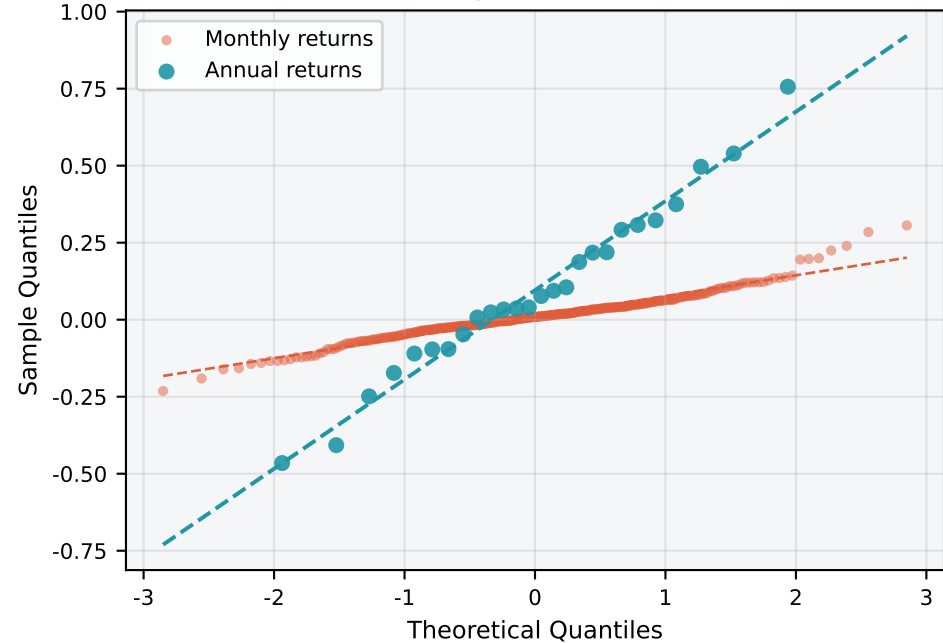
Historical Annual Returns (bars) vs Normal Fit (dashed)



Korea CPI Inflation Time Series (2001-2025)



Q-Q Plot: Monthly vs Annual KOSPI Returns
(closer to diagonal = more normal)

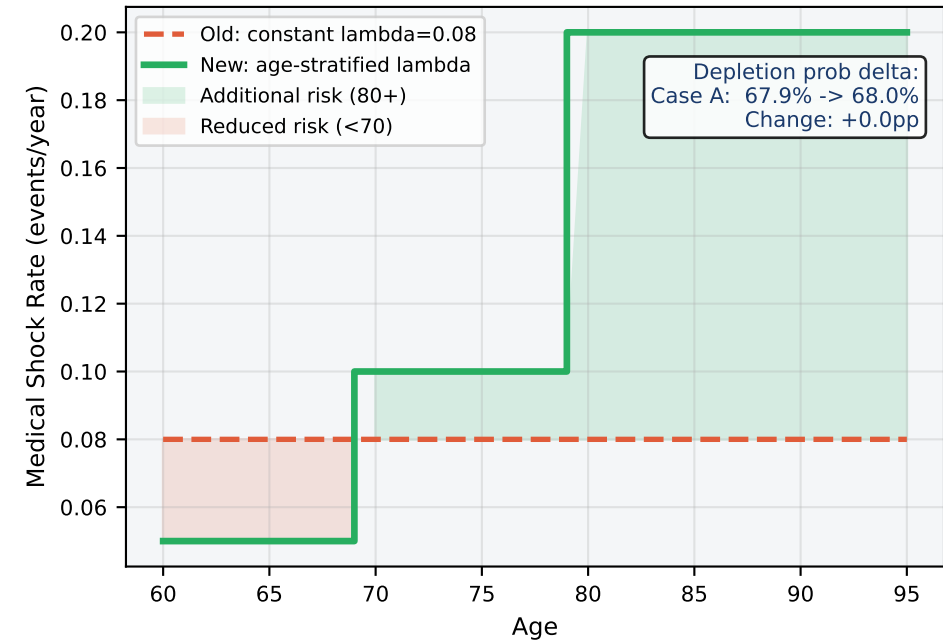


Normality Tests — Monthly vs Annual

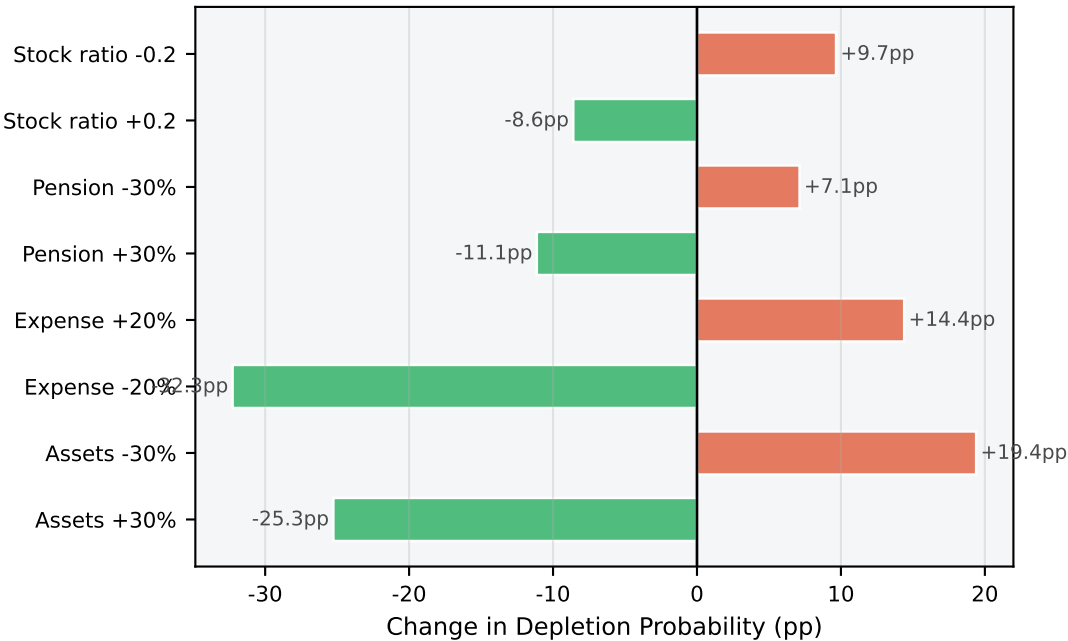
Metric	Monthly	Annual	Threshold
N observations	317	26	—
Mean	0.936%	9.559%	—
Std Dev	6.829%	27.484%	—
Excess Kurtosis	2.56	0.07	0.0 (normal)
Skewness	0.42	0.19	0.0 (normal)
Shapiro-Wilk p	0.496	0.958	> 0.05 = normal
Jarque-Bera p	0.000	0.920	> 0.05 = normal
Normality	REJECTED	NOT REJECTED	—

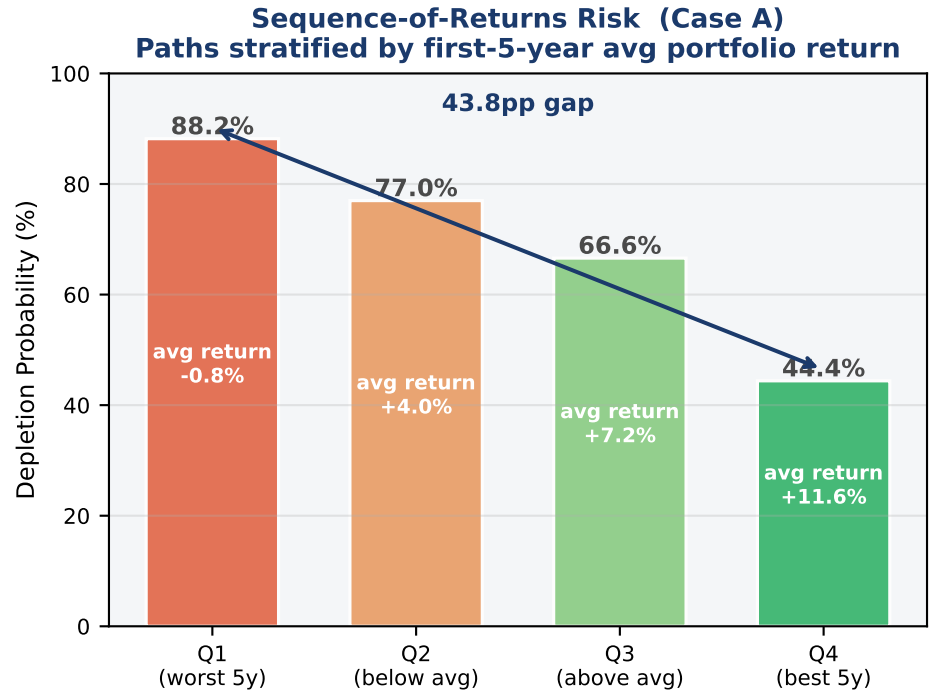
Simulation uses ANNUAL returns -> normality holds by CLT.

Medical Cost Lambda: Before vs After Fix
(NHIS: medical spending ~2x per decade after 65)



Sensitivity Analysis
(Base depletion: 68.6% | Case A, 1,500 paths)





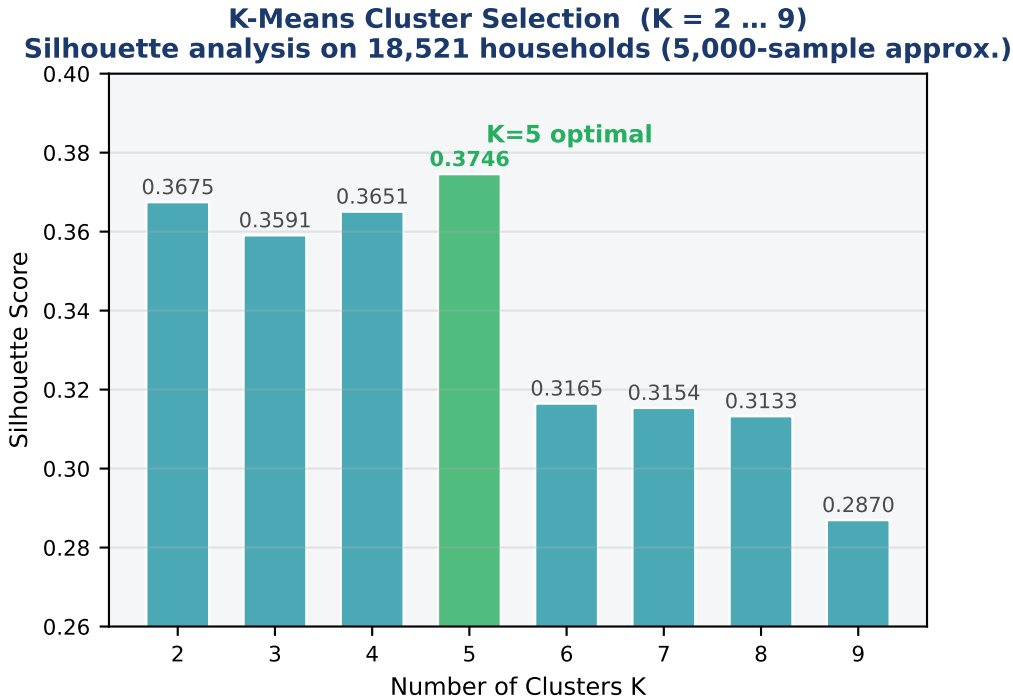
Why Sequencing Risk Matters

What it shows
All four quartiles are drawn from the SAME underlying return distribution (Normal, mean=9.6%, sd=28.0%). Only the ORDER of returns differs across paths.

The mechanism
Early bad returns reduce the capital base before recovery. With lower assets, subsequent withdrawals consume a higher FRACTION of remaining wealth — accelerating depletion. Conversely, strong early returns build a buffer that absorbs later downturns.

Implication for our model
Monte Carlo captures this effect implicitly: every path independently samples its own return sequence. The full depletion probability (e.g., 65%) is the expectation over ALL possible sequences, including adversely ordered ones. The fan chart width reflects sequencing uncertainty.

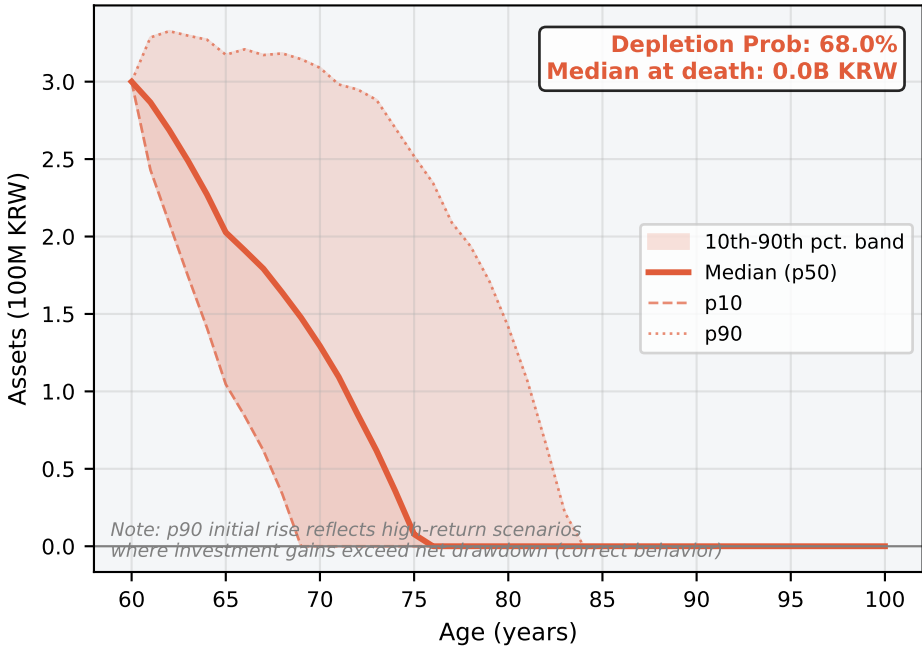
Key finding
Q1 paths (avg early return -0.8%): 88.2% depletion
Q4 paths (avg early return +11.6%): 44.4% depletion
Gap = 43.8pp — sequence dominates long-run outcome.



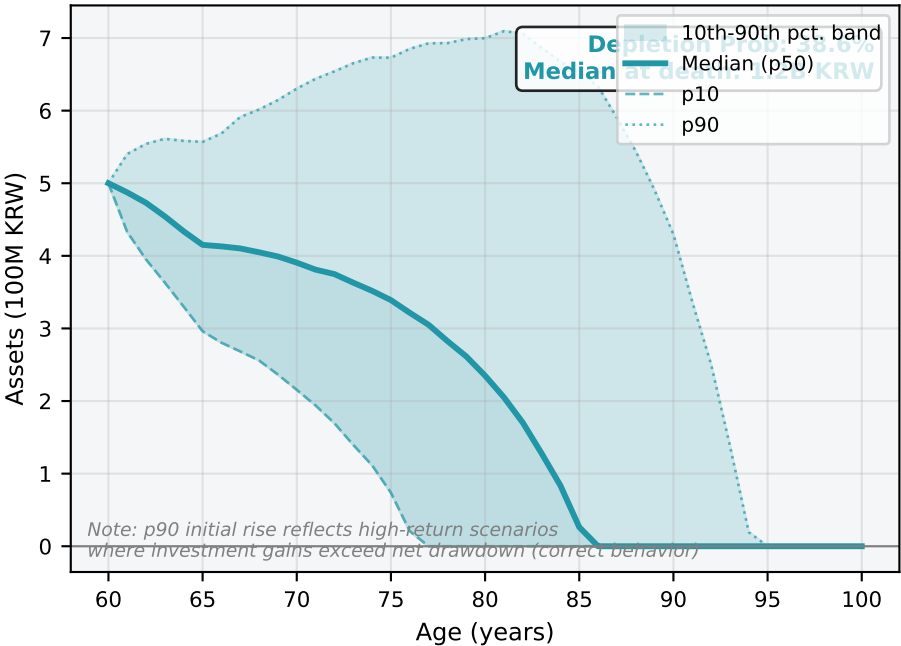
K-Selection: Results & Interpretation

K	Silhouette	Interpretation
2	0.3675	Splits only rich vs. rest — too c
3	0.3591	Adds middle class; misses asse
4	0.3651	Close to K=5 but conflates top
5 *	0.3746	Optimal — separates 5 wea
6	0.3165	Over-segments; ultra-rich clust
7	0.3154	Further fragmentation, lower c
8	0.3133	Diminishing returns on cluster
9	0.2870	Excessive — many near-empty

Case A: Male 60, Assets 300M KRW
Expense 250K/mo, Pension 100K/mo, w=0.40



Case B: Female 60, Assets 500M KRW
Expense 300K/mo, Pension 150K/mo, w=0.30



Depletion Age Distribution
(dashed = median depletion age)



Asset Path Summary Table (100M KRW)

Age	Case A p1	Case A p5	Case A p9	Case B p1	Case B p5	Case B p9
65y	1.05	2.03	3.17	2.96	4.15	5.57
70y	0.00	1.29	3.09	2.16	3.91	6.30
75y	0.00	0.08	2.52	0.74	3.39	6.73
80y	0.00	0.00	1.42	0.00	2.35	7.00
85y	0.00	0.00	0.00	0.00	0.26	6.53
90y	0.00	0.00	0.00	0.00	0.00	4.30
Depletion	68.0%	—	—	38.6%	—	—

A Cox PH Independence Assumption (HR multiplication)

Assumption: $HR_{total} = HR_{smoke} \times HR_{chronic} \times HR_{BMI}$ implies full independence between covariates.

Concern: Smoking and chronic disease are positively correlated. Joint HR may be overestimated for multi-risk profiles.

Rebuttal: Published HRs sourced from epidemiological meta-analyses that control for confounders via multivariable regression. The individual HRs are themselves adjusted estimates, partially accounting for covariate correlation.

Residual risk: Residual confounding remains. Estimated overestimation: ~10-20% for smoke+chronic combined profile (HR=2.03 vs plausible ~1.70-1.85).

Mitigation: Ideally: fit Cox PH from individual-level censored survival data (NHIS microdata). Unavailable for this project. Adding interaction term $H_{smoke} \times H_{chronic}$ would require empirical joint HR estimates from Korean cohort studies.

B Population Life Table vs Individual-Level Survival Data

Assumption: Baseline hazard $S_0(t)$ derived from aggregate national life table, not from individual censored survival records.

Justification: NHIS individual-level survival microdata with censoring indicators is not publicly available. Life table provides the only viable baseline.

Limitation: All individuals share the same baseline hazard function $S_0(t)$. Heterogeneity within age/sex strata (socioeconomic, genetic) is unmodeled.

Impact: Cox HR adjustment partially individualizes the curve. Residual heterogeneity causes predictions to regress toward population mean.

Note on median vs mean: $E[T]$ from life table = 23.2y => mean death age 83.2y. Simulated median ~85y gives $Mode(\sim 88) > Median(85) > Mean(83.2)$: LEFT-skewed. Mechanism: the baseline hazard $h(t)$ accelerates sharply in late life (Gompertz-like), concentrating deaths near the modal age (~88). In the early conditional interval (age 60-75), $h(t)$ remains low— few deaths occur here, but they extend the distribution leftward, forming a long thin left tail relative to the compressed right side. The hard upper bound at ~100 further truncates the right tail. Both effects together produce left-skewness: $Mean < Median$, consistent with our simulation. (Right-skewed would imply $Mean > Median$ — contradicted by data.)

C Normal Distribution for Equity Returns

Concern raised: KOSPI monthly returns exhibit fat tails. Normal distribution may underestimate tail-loss probability.

Empirical rebuttal: Simulation operates at ANNUAL frequency. Annual KOSPI returns: excess kurtosis = 0.07 (normal = 0.0), skewness = 0.19, Shapiro-Wilk $p = 0.96 >> 0.05$. Normality NOT rejected at annual level.

CLT argument: Aggregating 12 monthly returns to annual via geometric product reduces kurtosis substantially (central limit theorem). Monthly excess kurtosis = 2.56 collapses to 0.07 annually.

Residual concern: Rare extreme years (e.g., -46.5% in 2000) remain in the empirical distribution. Normal fit still samples plausible but bounded extreme values.

Alternative considered: t-distribution fit yields $df=35.1$ — effectively indistinguishable from Normal at this sample size. Historical simulation would require IID assumption on annual blocks.

D Medical Cost Shock Model (lambda fix applied in this version)

Previous version: Constant $\lambda = 0.08/\text{year}$ for all ages — one major shock every ~12.5 years regardless of age.

Problem: 80-year-olds face the same modeled shock frequency as 65-year-olds. NHIS data: per-capita medical spending roughly doubles each decade after 65.

Current version: Age-stratified λ : age < 70 -> 0.05 (1 per 20y), 70-79 -> 0.10 (1 per 10y), 80+ -> 0.20 (1 per 5y).

Impact measured: Case A depletion probability: 67.9% (old) -> 68.0% (new). $\Delta = +0.0pp$. More conservative and empirically grounded.

Remaining limitation: Shock magnitude (mean 500, SD 300, 10k KRW) is constant across age. In reality, catastrophic late-life care costs can exceed 10M KRW/event. Future: age-varying LogNormal parameters.

E Correlation Stationarity & Portfolio Constraint

Assumption: KOSPI-KTB correlation treated as constant at -0.025 (2009-2025).

Known limitation: Correlation regimes shift. In inflationary periods (2022-2023), equity-bond correlation turned positive in many markets.

Sensitivity: Sensitivity analysis shows stock ratio change of +/-0.2 moves depletion probability by 8.6 / 9.7 pp — moderate impact.

2-asset limitation: CVaR optimization with only KOSPI + KTB is mathematically degenerate: the feasible set is a single line segment w in $[0,1]$, so the "efficient frontier" collapses to a scalar weight. A simple target-return solve would yield identical results. We do NOT claim this as a core differentiator of the current system.

Actual role: The CVaR module serves as a principled risk-profiling layer: user risk tolerance (conservative/moderate/aggressive) -> optimal KOSPI weight -> Monte Carlo stock_ratio. This integration chain is functionally correct and non-trivial for the UX. CVaR provides downside-risk framing that variance-based methods miss.

Future value: With 5+ assets (domestic equity, foreign equity, REITs, IG bonds, alternatives), CVaR optimization becomes genuinely non-trivial: the feasible set is an N-1 dimensional simplex and Rockafellar-Uryasev linear reformulation provides significant computational advantage over MVO. Architecture is already extensible — only DataLoader expansion required.

AR(1) plan: CPI and KTB yields exhibit positive serial correlation (AR(1) structure visible in time series plot, p.3). Current model samples inflation i.i.d. each year — understates persistence in high-inflation regimes. Future: VAR(1) joint model for CPI + yield dynamics. Expected effect: wider fan chart spread and higher depletion prob under sustained inflation scenarios.

F Anomaly Detection Evaluation — Synthetic Test Data

Validation setup: F1-score (0.952) and AUC-ROC (1.000) reported for Layer 1 (transaction anomaly detection) were computed on SYNTHETIC test data, not on real labeled fraud records. This must be stated explicitly to avoid misleading evaluation.

Why synthetic: Real Korean financial fraud labels (ground-truth tagged transactions) are not publicly available. No open dataset with Korean banking transaction fraud labels exists for academic/competition use. Synthetic cases were designed to reflect known fraud patterns (voice phishing: new account + nighttime + consecutive + large amount).

Test set design: Normal: 30 small-amount daytime payments (3-40 10k KRW). Boundary: 10 moderate transfers to existing accounts (50-120 10k KRW). Anomaly-high: 20 large transfers (300-800 10k KRW). Anomaly-phishing: 20 nighttime new-account consecutive transfers. Boundary cases deliberately included to avoid inflated metrics.

Honest interpretation: The high AUC reflects that the engineered features (new account flag, hour-of-day, consecutive count) are highly discriminative for the synthetic patterns tested. Real-world performance against sophisticated fraud will be lower. Proper validation requires labeled transaction data from a financial institution.

Layer 3 caveat: Life pattern detection (Layer 3) uses entirely synthetic daily activity signals. No real elderly monitoring data was available. Threshold values (card transactions, conversation counts) are assumptions, not empirically calibrated.

Core Design: Bridging Probabilistic Rigor and Human Understanding**A STATISTICAL ENGINE OUTPUT**

Profile: Age 72 | Male | Chronic dx
Assets: 200M KRW | Expense: 220K/mo
Pension: 100K/mo | Non-smoker

Tool: run_monte_carlo()

$P(\text{depletion} < 85y) = 0.68$
median_depletion_age = 81
median_final_assets = 85M KRW

Tool: run_clustering()

cluster = 3 (mid-wealth elderly)
peer_wealth_percentile = 34

B AGENT TRANSFORMATION

Age Segment: 3 (≥ 75 years old)
Prompt: warm + concise + 1 core message

Language Rules Applied:

0.68 -> "열 번 중 일곱 번"
81y -> "여든한 살 즈음"
34th -> "100명 중 34번째"

Tool-call logic:

"얼마나 버티나요?"
--> run_monte_carlo(profile)

Parameters carried from context

C AGENT OUTPUT (Slow Banking UI)

User (age 72):
"제 돈이 얼마나 버틸까요?"

Agent:
"어르신, 컴퓨터가 계산을 해보니
지금처럼 생활하시면 여든한 살
즈음에는 주머니 사정이 조금
가벼워지실 수 있어요.

열 번 중에 일곱 번 정도는
돈이 먼저 마를 수 있다고 해요.

걱정 마세요. 연금을 더 유리하게
타는 방법을 알려드릴게요."

False Positive Handling & Multi-Turn Context Continuity**D ANOMALY DETECTION: FALSE POSITIVE HANDLING**

Trigger: Z-score 2.31 | New account | 23:15
350K KRW (3x personal baseline) | score: 0.847

Design: NO abrupt block.
Elderly confusion > transaction security.
False positive cost << false negative cost.

Agent (Slow Banking confirmation flow):

"어르신, 잠깐만요! 평소 안 보내시던
계좌에 큰 금액이에요.
본인이 직접 하신 거 맞나요?"

[YES] -> proceed + 24h review flag

[NO / hesitate / 30s no-response]

-> Hold + guardian alert + branch call

E MULTI-TURN CONTEXT CONTINUITY

Scenario: User refines conditions iteratively

Turn 1: "지출을 200만원으로 줄이면요?"
expense: 250K -> 200K (rest unchanged)
Depletion: 68% -> 55% (-13pp)

Turn 2: "담배도 끊으면요?"
smoking: True->False | HR: 2.03->1.35
E[death]: 83.2y -> 84.8y
Depletion: 55% -> 49% (-6pp)

Turn 3: "주식을 줄이면 더 안전해요?"
run_portfolio(conservative) -> KOSPI 12%
Depletion: 49% -> 46% (-3pp)

Agent never re-asks confirmed parameters.